Non-Maxwellian electron velocity distribution reconstruction from optical Thomson scattering using large neural networks

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Optical Thomson Scattering (OTS) is a critical diagnostic tool for plasma characterization. Traditional OTS diagnostics typically assume plasma equilibrium. However, in laser-driven inertial confinement fusion (ICF), the equilibrium hypothesis often becomes invalid due to laser-plasma instabilities and the generation of fast electrons. We have developed a reliable method for reconstructing arbitrary electron velocity distribution functions (EVDFs) using a large neural network that operates on the Fourier transform of the EVDF. Unlike traditional reconstruction algorithms, our approach makes no prior assumptions about the form of the EVDF. Moreover, the reconstruction requires only a single inference pass, enabling online, real-time OTS diagnostics in the future. We validated the reliability of this method with two cases: inverse Bremsstrahlung and superthermal electron generation by stimulated Raman scattering.

Optical Thomson scattering (OTS) provides a direct observation of electron motion in a plasma by encoding the electron velocities on the frequency spectrum of the scattered light. Owing to its precision, localized, and non-perturbative characters, OTS has enabled numerous important and successful applications in measuring plasma at extreme states, especially inertial confinement fusion (ICF)[1–3]. However, various mechanisms in laser-driven ICF—such as inverse Bremsstrahlung (IB)[4] radiation and laser plasma instabilities (LPI)[5]—cause the electron velocity distribution function (EVDF) to deviate significantly from a Maxwellian function (or generate superthermal electrons) [6], which departure from the Maxwellian EVDF assumption for conventional OTS[7].

Recently, OTS has been explored for inferring the non-Maxwellian EVDF in Langdon effect[8, 9], nonlocal heat transportation[10] and two-streams plasmas[11]. However, these investigations have consistently relied on assuming a specific functional form for the distribution, informed either by theory or simulations, which limits their applicability to truly unknown or complex non-equilibrium states. There were also approaches, such as angularly resolved OTS[12], aim to synergize the advantages of both non-collective and collective regimes to measure the EVDF over a broad dynamic range, and applying machine learning, such as transfer learning[13] and convolutional neural networks(CNNs)[14, 15], to resolve equilibrium plasmas from collective OTS spectra. Despite these significant advances, the challenge of achieving a precise, assumption-free inversion of the scattering spectrum into a general EVDF, without encountering solution ambiguities, remains an open problem.

We suggest that full reconstruction of EVDF from OTS spectrum can be achieved by improving the data inverting process with advenced neural networks technique. Two challenges were solved in this Letter. First, the forward mapping from the EVDF to the OTS spectrum is inherently discontinuous, where minute changes in the distribution function can lead to large variations in the simulated spectrum, making traditional inversion ill-posed. Second, iterative optimization approaches are highly susceptible to convergence at non-physical local minima, failing to recover the true global solution. To resolve the discontinuity issue, we represent the EVDF by its Fourier transform, which pushes all discontinuities to infinitely high wavenumbers; applying a low-pass constraint in the Fourier domain thus regularizes the mapping into a continuous function. To circumvent the pitfalls of conventional time-consuming iterative inversion—which often relies on gradient descent—we adopt a deep neural network (DNN) that directly maps the OTS spectrum to the EVDF, thereby enabling a single forward pass at inference time. Benefiting from the favorable over-parameterization of modern neural networks, we find that increasing model capacity readily yields a high-performance model with excellent generalization. We rigorously validated this generalization capability against published experimental data featuring inverse Bremsstrahlung effects and Particle-in-Cell (PIC) simulation data containing superthermal electrons. The method achieves a reconstruction error of less than 5%, and a single inference completes in approximately 1.5 ms (on personal computor@Intel(R) Core(TM) i5-8350U), demonstrating both high fidelity and practical utility for rapid diagnostic analysis.

The power of Thomson-scattered light with wave vector k_s in solid angle $d\Omega$ and in the frequency range from ω_s to $\omega_s + d\omega$ is[7, 16],

$$P_s d\Omega d\omega = \frac{1}{2\pi} I_0 r_e^2 V n_e \left(1 + \frac{2\omega_s}{\omega_i}\right) \times \left(\hat{k}_s \times \left(\hat{k}_s \times \hat{E}_i\right)\right)^2 S\left(\vec{k}_i - \vec{k}_s, \omega_i - \omega_s\right) d\Omega d\omega, \quad (1)$$

where r_e is the classical electron radius, V is the scattering volume, n_e is the electron number density, I_0 is the intensity of the probe light. $S(\vec{k}, \omega)$ is the spectral density function for a collisionless plasma,

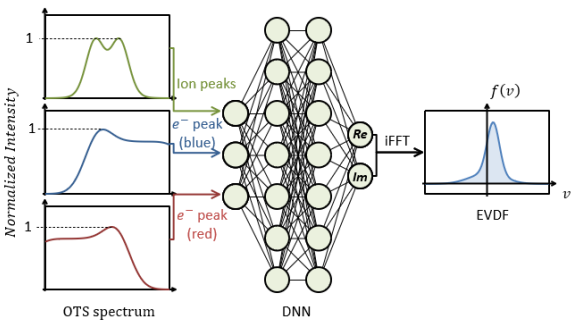


FIG. 1: illustrates the neural network-based reconstruction scheme. A deep neural network (DNN) is trained to predict the EVDF. The input is a 1,224-dimensional vector formed by concatenating three normalized spectral segments (ion: 349 - 353 nm, with 579 sampled points; red-shifted electron: 375 - 500 nm, with 304 sampled points; blue-shifted electron: 200 - 340 nm, with 341 sampled points), each normalized to a maximum of one. The network architecture includes a 1,224-dimensional input layer, two fully connected hidden layers of 5,000 dimensions each with leaky ReLU activation, and a 160-dimensional output layer (no activation). The output represents the Fourier transform of the EVDF (first 80 real, latter 80 imaginary parts), with the inverse transform yielding the EVDF over a velocity range of (-0.8 c,0.8c).

$$S\left(\vec{k},\omega\right) = \frac{2\pi}{k}\left(1-\frac{\chi_e}{\epsilon}\right)^2 f_e\left(\frac{\omega}{k}\right) + \sum_j \frac{2\pi}{k}\frac{p_j Z_j^2}{Z}\left(\frac{\chi_e}{\epsilon}\right) f_j\left(\frac{\omega}{k}\right), \quad (2)$$

where $\epsilon(\vec{k},\omega) = 1+\chi_e(\vec{k},\omega)+\sum_j \chi_j(\vec{k},\omega)$ represents the plasma dielectric permittivity. The probed wave vector (frequency) is the difference between the incident and scattered wave vectors (frequencies), $\vec{k} = \vec{k_i} - \vec{k_s}(\omega = \omega_i - \omega_s)$. We assume that the ion velocity distribution is Maxwellian, while the one-dimensional electron distribution function is obtained by integrating the full three-dimensional arbitrary velocity distribution function over the directions perpendicular to the probe direction. The summation is over all ion species j where p_j and Z_j are the fraction and ionization state of each. The electron velocity distribution function (EVDF), $f_e(v)$ is an arbitrary function which factors χ_e as well,

$$\chi_e\left(\vec{k},\omega\right) = \frac{n_e e^2}{m_e}\frac{4\pi}{k^2}\int dv\ \frac{\vec{k}\cdot\frac{\partial}{\partial\vec{v}}\left(f_e\left(v\right)\right)}{\vec{v}\cdot\vec{k}-\omega}, \quad (3)$$

where n_e, e and m_e are the number density, charge and mass of electrons. The integral is over the 3-dimension velocity space.

Figure 1 shows the architecture of DNN model to invert EVDF from OTS spectra. The input spectrum to the DNN is divided into three segments: two segments for electron peaks (375–500 nm and 200–340 nm) and one segments for ion peaks (349–353 nm). In experiments, the electron and ion peaks differ dramatically in both intensity and spectral resolution—by roughly two orders of magnitude each. Because of the huge difference in intensity, we normalize each segment to its maximum ensures that the weak electron peaks is not overwhelmed by the strong ion signal during reconstruction. The DNN outputs the real and imaginary parts of the Fourier transform of EVDFs. The real part represents the component symmetric about zero velocity, while the imaginary part represents the anti-symmetric component. After combining the real and imaginary parts into a full complex array, we apply an inverse Fourier transform (using NumPy's ifft function) to recover the EVDF. The loss function is defined as the mean squared error of the difference between the EVDF predicted from the OTS spectra and the ground truth in the training set. The Adam optimizer is used for the training process, which is divided into four phases: the first 3,000 epochs are performed with a learning rate of 1×10^{-4}, followed by 6,000 epochs with a reduced learning rate of 1×10^{-5}, and then 10,000 epochs with 1×10^{-6}, and finally 100,000 epochs with 1×10^{-7}. The DNN was trained on a cpu node (2 x Intel Xeon Scalable Cascade Lake 6248(2.5GHz, 20 cores)) with around 10 hours to complete.

Figure 2(a) illustrates the performance of neural networks with different architectures tested in our study. Based on the statistical results presented in the figure, we provided an empirical formula that estimates the optimal performance achievable for a given number of parameters, N,

$$L = \frac{0.3678}{N^{0.2685}}, \quad (4)$$

where L is the average loss of neural networks on the training dataset. Furthermore in fig. 2(a), we found that the most commonly used convolutional neural networks (CNNs) [15] performed poorly on this specific problem. This is because CNN architectures inherently assume translational and rotational invariance within the data [ref], properties which OTS spectra do not possess. In

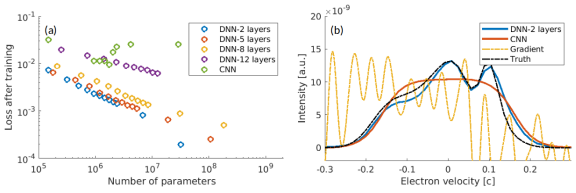


FIG. 2: (a) shows the loss value after training of different neural networks, including CNNs and DNNs with different number of hidden layers(2, 5, 8 and 12). (b) compares reconstructed EVDFs using different methods (DNN, CNN, gradient descent).

our experiments, we found that increasing the number of hidden layers in DNNs did not improve performance for our problem. In fact, when the number of hidden layers exceeded eight, the performance droped with further increases in depth. While it is commonly believed that deeper networks enhance generalization[ref], our tests showed that a DNN with just two hidden layers already achieved sufficiently good generalization for EVDF reconstructions, and adding more layers caused a significant drop in performance due to vanishing gradients[ref]. Figure 2(b) also compares the reconstruction effectiveness of the gradient descent method. In the gradient descent approach, the EVDF was treated as the adjustable parameters. However, due to the abundance of local minima in this specific inversion problem, it always failed to converge to the global optimum solution.

The dataset is generated by randomly producing non-equilibrium velocity distributions and calculating their corresponding OTS spectra based on the theory mentioned previously. Each EVDF is constructed by superimposing 10 randomly generated Gaussian functions. The width, amplitude, and center position of each Gaussian are uniformly sampled from predefined ranges (width from 0.02 to 0.08 c, ampliture from 0.5 to 1.5, center from -0.2 to 0.2 c). After superposition, the resulting velocity distribution is normalized. Both the training set (5,000 spectra) and the test set (100 spectra) are generated using this method. To improve the efficiency of spectra computation, we propose an equivalent formula,

$$\chi_e\left(\vec{k},\omega\right) = \frac{n_e e^2}{i m_e}\frac{4\pi}{k^2}\int_0^\infty dK K F_e\left(K\right) exp\left(-i\frac{K}{k}\omega\right), \tag{5}$$

where $F_e(K)$ is the Fourier transform of the velocity distribution which is $F_e(K) = \int_{-\infty}^{+\infty} dK f_e(v) exp(iKv)$. With the Fourier expansion method, we only require 80 components to compute a spectrum with a relative error nearly 0.015% reducing the computation time by 1,000 times(check!). The entire dataset and test set can be generated within approximately 800 seconds. The probe laser wavelength and observation position are specified. The plasma ions consist of CH, with the relative concentration of C and H, as well as the ionization degree of C, being randomly generated. Additionally, C and H ions are assigned random ion temperatures and drift velocities. Because the much larger ion mass strongly suppresses their laser-driven deviations from equilibrium, and for simplicity as commonly assumed in the literature[ref], we take the ion velocity distributions to remain Maxwellian. Local charge neutrality is assumed throughout the plasma. Thus, the neural network model enables the simultaneous inversion of all key plasma properties.

After training, the model achieves a final loss of 1.68×10^{-4} on the training set. Figure 3(a) and (b) presents selected reconstruction examples from the test set; complete results are provided in the Supplementary Material. Figure 3(c) and (d) shows reconstructions of super-Gaussian EVDFs, which were used to evaluate model generalization (as such functions are absent from the training set) and practical relevance, given that super-Gaussian distributions arise from inverse Bremsstrahlung in plasma experiments.

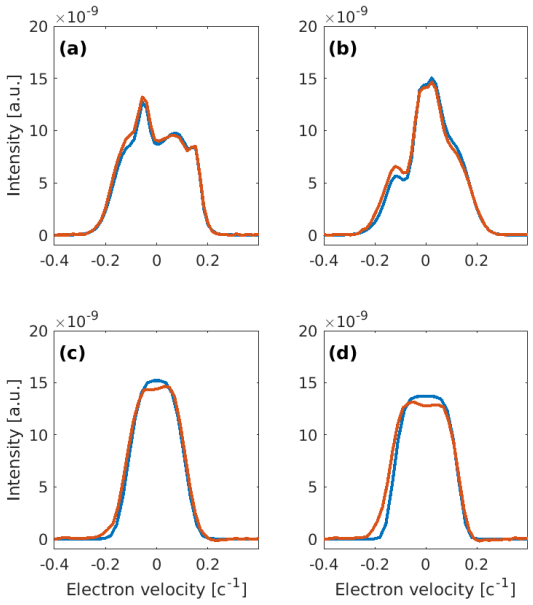


FIG. 3: shows reconstructed non-Maxwellian EVDFs. Blue curves are truth EVDFs and red curves are reconstructed EVDFs. (a) and (b) are EVDFs from the test set. (c) and (d) are Super-Gaussian EVDFs. The super-Gaussian coefficients are 3.5 and 5.

We focus on using the proposed method to diagnose hot electrons in laser-produced plasmas. Due to laser–plasma instabilities, plasma waves with longitudinal electric fields can form. When some electrons move in the same direction as the phase velocity of these waves and at a comparable speed, they are accelerated to higher energies, becoming hot electrons. Existing techniques can only diagnose hot electrons ex-situ, after they have left the plasma. In contrast, laser Thomson scattering offers the potential for in-situ diagnosis. Specifically, when an electron moves parallel to the laser propagation direction, its Thomson-scattered light undergoes a Doppler shift proportional to its velocity. Because hot electrons move much faster than background electrons, they produce distinct redshifted and blueshifted signals at the far ends of the spectrum, which can be used to reconstruct their properties.

We performed PIC simulations to study hot electrons generated by stimulated Raman scattering (SRS) using the following parameters. In these simulations, the number density of hot electrons is about 0.1–1% of the background electron density. To capture this feature, we augmented the original training set of EVDFs by adding a weak random Gaussian distribution to simulate hot electrons, with the height, width, and center position randomly chosen within specified ranges. Figure 4(a) com-

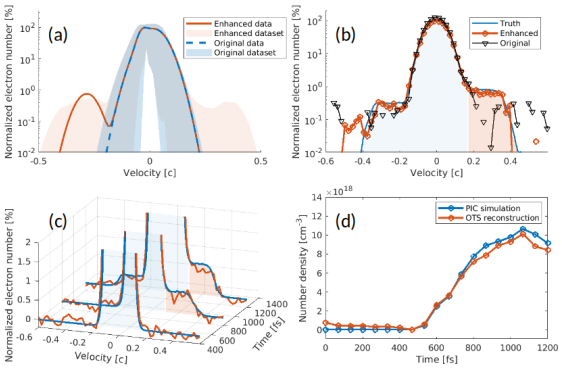


FIG. 4: shows reconstruction of EVDFs containing hot electrons. The ordinates in (a–c) represent normalized electron number density, with the maximum value of the velocity distribution normalized to unity. (a) illustrates a sample of training EVDF before (blue curve) and after (red curve) enhancement. Shadow areas mark the average EVDF of all samples in training set before (blue) and after (red) enhancement. (b) compares the PIC simulated EVDF (blue curve) with reconstructed EVDFs by DNNs trained on datasets with (red curve) and without (black curve) hancement. (c) shows the reconstruction EVDFs (red curves) of the DNN trained on the enhanced database at different time, compared with simulated EVDFs (blue curves). Hot electrons are marked by red shadow areas. (d) presents the temporal evolution of forward-propagating hot electrons. Blue and red curves are from PIC simulation and OTS reconstruction seperately.

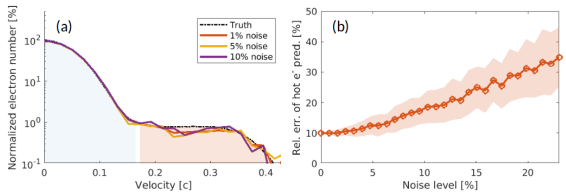


FIG. 5: investigates the robustness of the reconstruction to spectral measurement noise. (a) compares EVDFs reconstructed from spectra with different noise levels (1%, 5%, 10%) and the ground-truth EVDF (from PIC simulation). (d) presents mean values (red curve) and standard deviations (red area) of relative errors in number of hot electron prediction at different noise levels. Values at each noise level were calculated from 700 iterations of different EVDFs and random noise.

pares EVDFs from the original and augmented datasets. As shown in Fig. 4(b), the model trained on the augmented dataset accurately reconstructs the hot-electron component. Notably, although only Gaussian distributions were used to represent hot electrons during training, the model generalizes this feature and successfully reconstructs both forward- and backward-propagating hot electrons (see Fig. 4(c)).

Figure 5 demonstrates the reconstruction performance in the presence of spectral measurement noise. We added varying levels of random noise to the OTS spectra and fed them directly into the trained model. The resulting errors in the reconstructed velocity distribution remain within acceptable bounds, indicating good robustness. This robustness arises because the training set contains only clean, noise-free OTS spectra. Unlike gradient-descent-based reconstruction, which indiscriminately fits every detail in the spectrum including noise, our neural network model, when artificial noise is added to the test spectra, tends to filter out noise-related features that are absent from the training set and relies instead on features that appear abundantly in the training data. Consequently, the neural network method possesses inherent robustness against measurement noise.

In summary, we demonstrate that a large-scale, Fourier-feature-based two-hidden-layer DNN can reliably reconstruct the EVDF from OTS spectra. The DNN model's generalization and practical utility are validated on inverse Bremsstrahlung and SRS superthermal electron examples. We find that shallow DNN models perform far better than CNN models and deep DNNs. The reconstruction quality can be improved simply by increasing the number of parameters. Contrary to the conventional belief that more parameters inevitably cause overfitting [ref], the DNN benefits from favorable over-parameterization [ref] on this problem, reminiscent of the scaling laws observed in large language models [ref]. Furthermore, DNN gains a decisive advantage over gradient descent by exploiting statistical regularities learned from the dataset, thus avoiding the local-optima traps that plague gradient-based optimization. We further demonstrate that a superposition of Gaussian functions offers an efficient and versatile way to generate EVDF datasets across different physical regimes, which is a key challenge in applying neural-network-based methods.

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